Piecewise-Defined & Implicit Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating piecewise functions

Given $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$, find $f(-3)$ and $f(4)$.

$f(-3)$: since $-3 < 0$, use x^2: $f(-3) = (-3)^2 = 9$. $f(4)$: since $4 \geq 0$, use $2x + 1$: $f(4) = 2(4) + 1 = 9$.

2 [Foundational] • Target: Evaluating piecewise functions

A shipping company charges a flat 5 dollars for packages up to 2 kg, plus 1.50 dollars per kg for anything over 2 kg. Write a piecewise function $C(w)$ for the cost, and find $C(5)$.

$C(w) = 5$ for $0 < w \leq 2$, and $C(w) = 5 + 1.5(w - 2)$ for $w > 2$. $C(5) = 5 + 1.5(5 - 2) = 5 + 4.5 = 9.5$, so 9.5 dollars.

3 [Foundational] • Target: Checking continuity of piecewise functions

Find the value of k that makes $f(x) = 3x + k$ for $x < 2$, and $f(x) = x^2 - 1$ for $x \geq 2$, continuous at $x = 2$.

For continuity, the two pieces must agree at $x = 2$: $3(2) + k = 2^2 - 1$, so $6 + k = 3$, giving $k = -3$.

4 [Foundational] • Target: Checking continuity of piecewise functions

Determine whether $g(x) = x + 1$ for $x \leq 1$, and $g(x) = 3x - 2$ for $x > 1$, is continuous at $x = 1$, showing your reasoning.

Left piece at $x = 1$: $1 + 1 = 2$. Right piece as $x \to 1^+$: $3(1) - 2 = 1$. Since $2 \neq 1$, the two pieces do not match at $x = 1$, so g is not continuous there — there is a jump discontinuity.

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5 [Foundational] • Target: Visualizing a piecewise graph

The graph shows $f(x) = -x - 1$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$.

a Evaluate the left piece and the right piece of f at $x = 0$, and use this to determine whether f is continuous there. Left piece as $x \to 0^-$: $-0 - 1 = -1$. Right piece at $x = 0$: $0^2 = 0$. Since $-1 \neq 0$, f has a jump discontinuity at $x = 0$, visible as the gap between the two branches in the graph.

b State the range of f shown on the graph. The linear branch for $x < 0$ takes all values greater than -1 (approaching but not including -1), and the parabola branch for $x \geq 0$ takes all values ≥ 0. Combined range: $(-1, \infty)$.

(a) Left piece as $x \to 0^-$: $-0 - 1 = -1$. Right piece at $x = 0$: $0^2 = 0$. Since $-1 \neq 0$, f has a jump discontinuity at $x = 0$, visible as the gap between the two branches in the graph.; (b) The linear branch for $x < 0$ takes all values greater than -1 (approaching but not including -1), and the parabola branch for $x \geq 0$ takes all values ≥ 0. Combined range: $(-1, \infty)$.

6 [Foundational] • Target: Implicit relations and testing for functions

Determine whether $x^2 + y^2 = 25$ defines y as a function of x. Justify using the vertical line test.

This is a circle of radius 5. A vertical line such as $x = 3$ intersects it at two points ($y = 4$ and $y = -4$), so it fails the vertical line test — y is not a function of x.

7 [Foundational] • Target: Implicit relations and testing for functions

Solve $x^2 + y^2 = 25$ for y explicitly, and state which branch corresponds to a function.

$y = \pm\sqrt{25 - x^2}$. Each branch individually is a function: $y = \sqrt{25 - x^2}$ (upper semicircle) or $y = -\sqrt{25 - x^2}$ (lower semicircle) — together they form the full circle, which is not a function.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Implicit differentiation-style reasoning (algebraic slope)

For the implicit relation $x^2 + 4y^2 = 16$ (an ellipse), find the two y-values when $x = 2$, and describe what this means about the relation as a function.

$4 + 4y^2 = 16$, so $4y^2 = 12$, $y^2 = 3$, giving $y = \pm\sqrt{3}$. Since $x = 2$ produces two distinct y-values, the relation is not a function of x over its full domain.

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9 [Intermediate] • Target: Modeling with piecewise functions

A tax schedule charges 10% on income up to 10{,}000 dollars and 20% on income above 10{,}000 dollars. Write a piecewise tax function $T(x)$ for income x, and find the tax owed on an income of 25{,}000 dollars.

$T(x) = 0.10x$ for $0 \le x \le 10000$, and $T(x) = 1000 + 0.20(x - 10000)$ for $x > 10000$.
$T(25000) = 1000 + 0.20(15000) = 1000 + 3000 = 4000$, so 4000 dollars in tax.

10 [Intermediate] • Target: Piecewise functions with three or more pieces

A parking garage charges 3 dollars for the first hour, 2 dollars for each additional hour up to 5 hours total, and a flat 15 dollars for anything over 5 hours. Write this as a three-piece function $P(h)$ for h hours parked, and find $P(4)$.

$P(h) = 3$ for $0 < h \le 1$; $P(h) = 3 + 2(h - 1)$ for $1 < h \le 5$; $P(h) = 15$ for $h > 5$.
$P(4) = 3 + 2(4 - 1) = 3 + 6 = 9$, so 9 dollars.

11 [Intermediate] • Target: Piecewise functions with three or more pieces

Evaluate the absolute value function $f(x) = |2x - 6|$ at $x = 1$ and $x = 5$ by first rewriting it as a two-piece function without absolute value bars.

Since $2x - 6 = 0$ at $x = 3$: $f(x) = -(2x - 6) = 6 - 2x$ for $x < 3$, and $f(x) = 2x - 6$ for $x \ge 3$.
$f(1) = 6 - 2(1) = 4$. $f(5) = 2(5) - 6 = 4$.

12 [Intermediate] • Target: Graphing implicit relations point-by-point

For the implicit relation $y^2 = x + 4$, find all y-values when $x = 0$, $x = 5$, and $x = -4$, and state the smallest allowed x-value.

At $x = 0$: $y^2 = 4$, so $y = \pm 2$. At $x = 5$: $y^2 = 9$, so $y = \pm 3$. At $x = -4$: $y^2 = 0$, so $y = 0$ (a single point). Since $y^2 \ge 0$ requires $x + 4 \ge 0$, the smallest allowed x-value is $x = -4$.

13 [Intermediate] • Target: Piecewise functions and average rate of change

For $f(x) = x^2$ on $x < 2$ and $f(x) = 3x - 2$ on $x \ge 2$, find the average rate of change of f on $[0, 4]$.

$f(0) = 0^2 = 0$ (using the first piece). $f(4) = 3(4) - 2 = 10$ (using the second piece, since $4 \ge 2$). AROC $= \frac{10 - 0}{4 - 0} = \frac{10}{4} = 2.5$.

14 [Intermediate] • Target: Piecewise functions and average rate of change

Given $g(x) = 2x + 1$ for $x \le 3$ and $g(x) = x^2 - 5$ for $x > 3$, find the average rate of change of g on $[1, 5]$.

$g(1) = 2(1) + 1 = 3$ (first piece, since $1 \le 3$). $g(5) = 5^2 - 5 = 20$ (second piece, since $5 > 3$). AROC $= \frac{20 - 3}{5 - 1} = \frac{17}{4} = 4.25$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Piecewise functions and average rate of change • Cross-Lesson: 2.02 Logarithmic Functions & Properties**

A cell phone plan charges 20 dollars flat for up to 2 GB of data, plus 8 dollars per GB for each additional GB, up to a maximum monthly charge of 68 dollars (unlimited data cap). Write a three-piece function $M(x)$ for the monthly cost based on x GB used, and find $M(8)$.

$M(x) = 20$ for $0 \le x \le 2$; $M(x) = 20 + 8(x - 2)$ for $2 < x \le 8$ (since $20 + 8(6) = 68$ is reached exactly at $x = 8$); $M(x) = 68$ for $x > 8$. $M(8) = 20 + 8(8 - 2) = 20 + 48 = 68$, so 68 dollars.

16 **[Challenging] • Target: Visualizing a piecewise graph • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle**

Find the value of k that makes $f(x) = 7x + k$ for $x < 0$, and $f(x) = 8x - 9$ for $x \ge 0$, continuous at $x = 0$.

Final Answer: -9. For continuity: $7(0) + k = 8(0) - 9$, so $0 + k = -9$, giving $k = -9$.

17 **[Challenging] • Target: Implicit relations and testing for functions • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs**

Find the value of k that makes $f(x) = 9x + k$ for $x < 3$, and $f(x) = -x - 2$ for $x \ge 3$, continuous at $x = 3$.

Final Answer: -32. For continuity: $9(3) + k = -1(3) - 2$, so $27 + k = -5$, giving $k = -32$.

18 **[Challenging] • Target: Implicit differentiation-style reasoning (algebraic slope) • Cross-Lesson: 4.01 Inverse & Composite Functions**

Find the value of k that makes $f(x) = -7x + k$ for $x < 3$, and $f(x) = 7x - 8$ for $x \ge 3$, continuous at $x = 3$.

Final Answer: 34. For continuity: $-7(3) + k = 7(3) - 8$, so $-21 + k = 13$, giving $k = 34$.

19 **[Challenging] • Target: Modeling with piecewise functions • Cross-Lesson: 4.03 Parametric Equations & Modeling**

Find the value of k that makes $f(x) = x + k$ for $x < -1$, and $f(x) = 8x + 3$ for $x \ge -1$, continuous at $x = -1$.

Final Answer: -4. For continuity: $1(-1) + k = 8(-1) + 3$, so $-1 + k = -5$, giving $k = -4$.

20 **[Challenging] • Target: Piecewise functions with three or more pieces • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros**

Find the value of k that makes $f(x) = 8x + k$ for $x < 5$, and $f(x) = -x + 5$ for $x \ge 5$, continuous at $x = 5$.

Final Answer: -40. For continuity: $8(5) + k = -1(5) + 5$, so $40 + k = 0$, giving $k = -40$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Evaluating piecewise functions

Given $f(x) = 3x - 5$ for $x < 0$, and $f(x) = -5x + 9$ for $x \geq 0$, find $f(-5)$.

a. 34

b. -19

c. -20 ✓

d. -21

Since $-5 < 0$, use $3x - 5$: $f(-5) = -20$.

22 [Foundational] • Target: Checking continuity of piecewise functions

Given $f(x) = -6x + 4$ for $x < 4$, and $f(x) = 6x + 5$ for $x \geq 4$, find $f(8)$.

a. 53 ✓

b. 54

c. -44

d. 52

Since $8 \geq 4$, use $6x + 5$: $f(8) = 53$.

23 [Foundational] • Target: Visualizing a piecewise graph

Given $f(x) = -3x + 3$ for $x < 1$, and $f(x) = -6x - 9$ for $x \geq 1$, find $f(1)$.

a. -15 ✓

b. -14

c. 0

d. -16

Since $1 \geq 1$, use $-6x - 9$: $f(1) = -15$.

24 [Foundational] • Target: Implicit relations and testing for functions

Given $f(x) = 4x - 7$ for $x < 2$, and $f(x) = 6x - 5$ for $x \geq 2$, find $f(3)$.

a. 14

b. 5

c. 13 ✓

d. 12

Since $3 \geq 2$, use $6x - 5$: $f(3) = 13$.

25 [Foundational] • Target: Implicit differentiation-style reasoning (algebraic slope)

Given $f(x) = 6x + 1$ for $x < -1$, and $f(x) = x - 2$ for $x \geq -1$, find $f(-5)$.

a. -29 ✓

b. -30

c. -28

d. -7

Since $-5 < -1$, use $6x + 1$: $f(-5) = -29$.

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26 [Foundational] • Target: Modeling with piecewise functions

Given $f(x) = 2x - 7$ for $x < -1$, and $f(x) = -5x - 4$ for $x \geq -1$, find $f(-6)$.

a. 26

b. -18

c. -19 ✓

d. -20

Since $-6 < -1$, use $2x - 7$: $f(-6) = -19$.

27 [Foundational] • Target: Piecewise functions with three or more pieces

Given $f(x) = -3x + 8$ for $x < -3$, and $f(x) = -2x + 6$ for $x \geq -3$, find $f(-7)$.

a. 28

b. 30

c. 29 ✓

d. 20

Since $-7 < -3$, use $-3x + 8$: $f(-7) = 29$.

28 [Foundational] • Target: Graphing implicit relations point-by-point

Given $f(x) = -2x + 2$ for $x < 4$, and $f(x) = -x - 4$ for $x \geq 4$, find $f(-1)$.

a. 4 ✓

b. 5

c. -3

d. 3

Since $-1 < 4$, use $-2x + 2$: $f(-1) = 4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Piecewise functions and average rate of change

Find the value of k that makes $f(x) = -2x + k$ for $x < 2$, and $f(x) = 7x - 1$ for $x \geq 2$, continuous at $x = 2$.

a. 17 ✓

b. 16

c. 9

d. 18

For continuity: $-2(2) + k = 7(2) - 1$, so $-4 + k = 13$, giving $k = 17$.

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30 [Intermediate] • Target: Evaluating Piecewise Functions

Find the value of k that makes $f(x) = -9x + k$ for $x < -2$, and $f(x) = -7x + 6$ for $x \geq -2$, continuous at $x = -2$.

a. 38

b. 2 ✓

c. 3

d. 1

For continuity: $-9(-2) + k = -7(-2) + 6$, so $18 + k = 20$, giving $k = 2$.

31 [Intermediate] • Target: Continuity of Piecewise Functions at Boundaries

Find the value of k that makes $f(x) = 3x + k$ for $x < 6$, and $f(x) = -5x - 8$ for $x \geq 6$, continuous at $x = 6$.

a. -57

b. -20

c. -55

d. -56 ✓

For continuity: $3(6) + k = -5(6) - 8$, so $18 + k = -38$, giving $k = -56$.

32 [Intermediate] • Target: Implicit Relations and Conic Sections

Find the value of k that makes $f(x) = -9x + k$ for $x < -4$, and $f(x) = 2x - 6$ for $x \geq -4$, continuous at $x = -4$.

a. -49

b. 22

c. -51

d. -50 ✓

For continuity: $-9(-4) + k = 2(-4) - 6$, so $36 + k = -14$, giving $k = -50$.

33 [Intermediate] • Target: Asymptotes and Branches of Hyperbolas

Find the value of k that makes $f(x) = -8x + k$ for $x < 5$, and $f(x) = -2x + 2$ for $x \geq 5$, continuous at $x = 5$.

a. 33

b. -48

c. 31

d. 32 ✓

For continuity: $-8(5) + k = -2(5) + 2$, so $-40 + k = -8$, giving $k = 32$.

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34 [Intermediate] • Target: Evaluating piecewise functions

For the implicit relation $x^2 + y^2 = 25$, find all y-values when $x = 3$.

a. $y = \pm 4$ ✓

b. $y = 4$

c. $y = \pm 3$

d. $y = \pm 5$

$3^2 + y^2 = 25$: $y^2 = 25 - 9 = 16$, so $y = \pm 4$.

35 [Intermediate] • Target: Checking continuity of piecewise functions

For the implicit relation $x^2 + y^2 = 1681$, find all y-values when $x = 9$.

a. $y = 40$

b. $y = \pm 40$ ✓

c. $y = \pm 41$

d. $y = \pm 9$

$9^2 + y^2 = 1681$: $y^2 = 1681 - 81 = 1600$, so $y = \pm 40$.

36 [Intermediate] • Target: Visualizing a piecewise graph

For $f(x) = 5x - 8$ on $x < 0$ and $f(x) = -3x + 7$ on $x \geq 0$, find the average rate of change of f on $[-1, 2]$.

a. $\frac{14}{3}$ ✓

b. $-\frac{14}{3}$

c. $\frac{11}{3}$

d. $\frac{17}{3}$

$f(-1) = -13$ (first piece). $f(2) = 1$ (second piece, since $2 \geq 0$). AROC $= \frac{1-(-13)}{2-(-1)} = \frac{14}{3}$.

37 [Intermediate] • Target: Implicit relations and testing for functions

For $f(x) = -x + 8$ on $x < -3$ and $f(x) = 6x + 7$ on $x \geq -3$, find the average rate of change of f on $[-5, 0]$.

a. $-\frac{6}{5}$ ✓

b. $-\frac{11}{5}$

c. $-\frac{1}{5}$

d. $\frac{6}{5}$

$f(-5) = 13$ (first piece). $f(0) = 7$ (second piece, since $0 \geq -3$). AROC $= \frac{7-(13)}{0-(-5)} = -\frac{6}{5}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Implicit differentiation-style reasoning (algebraic slope) • Cross-Lesson: 4.01 Inverse & Composite Functions

A shipping company charges a flat 7 dollars for packages up to 3 kg, plus $\frac{5}{2}$ dollars per kg for anything over 3 kg. Find the cost for a 13 kg package.

a. 32 ✓

b. 33

c. 31

d. $\frac{79}{2}$

$C(13) = 7 + \frac{5}{2}(13 - 3) = 7 + \frac{5}{2}(10) = 32$ dollars.

39 [Challenging] • Target: Modeling with piecewise functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

A shipping company charges a flat 13 dollars for packages up to 2 kg, plus 2 dollars per kg for anything over 2 kg. Find the cost for a 7 kg package.

a. 27

b. 24

c. 23 ✓

d. 22

$C(7) = 13 + 2(7 - 2) = 13 + 2(5) = 23$ dollars.

40 [Challenging] • Target: Piecewise functions with three or more pieces • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A shipping company charges a flat 8 dollars for packages up to 2 kg, plus 3 dollars per kg for anything over 2 kg. Find the cost for a 9 kg package.

a. 29 ✓

b. 30

c. 28

d. 35

$C(9) = 8 + 3(9 - 2) = 8 + 3(7) = 29$ dollars.

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41 [Challenging] • Target: Evaluating Piecewise Functions • Cross-Lesson: 1.02 Rational Functions & Asymptotes

A tax bracket levies no tax on the first 10,000 dollars, and 15% on income exceeding 10,000 dollars. Which piecewise function $T(x)$ models total tax for income $x \geq 0$?

a. $T(x) = 0$ **for** $x \leq 10000$**, and** $T(x) = 0.15(x - 10000)$ **for** $x > 10000$ ✓

b. $T(x) = 0.15x$ for all x

c. $T(x) = 0$ for $x \leq 10000$, and $T(x) = 0.15x$ for $x > 10000$

d. $T(x) = 0$ for $x \leq 10000$, and $T(x) = 1500 + 0.15x$ for $x > 10000$

No tax is paid on income up to 10,000 ($T = 0$). Only the amount beyond 10,000 (($x - 10000$)) is taxed at 15%, yielding $T(x) = 0.15(x - 10000)$.

42 [Challenging] • Target: Continuity of Piecewise Functions at Boundaries • Cross-Lesson: 1.03 Rates of Change & Concavity

Find the value of k that makes $f(x)$ continuous at $x = 3$, where $f(x) = kx + 1$ for $x \leq 3$ and $f(x) = 2x^2 - 5$ for $x > 3$.

a. 4 ✓

b. 2

c. 5

d. 13

For continuity at $x = 3$, $\lim_{x \to 3^-} f(x) = \lim_{x \to 3^+} f(x)$. Left limit: $k(3) + 1 = 3k + 1$. Right limit: $2(3^2) - 5 = 18 - 5 = 13$. Setting $3k + 1 = 13 \Rightarrow 3k = 12 \Rightarrow k = 4$.

43 [Challenging] • Target: Continuity of Piecewise Functions at Boundaries • Cross-Lesson: 2.01 Exponential Functions & Growth Models

A function h is defined by $h(x) = |x - 2| + 3$. Where is $h(x)$ continuous and where is it not differentiable?

a. **Continuous everywhere; not differentiable at** $x = 2$ ✓

b. Continuous only for $x \geq 2$; differentiable everywhere

c. Discontinuous at $x = 2$; differentiable elsewhere

d. Continuous and differentiable everywhere on $(-\infty, \infty)$

Absolute value functions are continuous everywhere. However, at $x = 2$ there is a sharp corner, where the left-hand slope (-1) differs from the right-hand slope ($+1$), so it is not differentiable at $x = 2$.

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44 [Challenging] • Target: Implicit Relations and Conic Sections • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Find the center and radius of the circle given by $x^2 + y^2 - 6x + 8y = 0$.

a. **Center** $(3, -4)$**, Radius** 5 ✓

b. Center $(-3, 4)$, Radius 5

c. Center $(3, -4)$, Radius 25

d. Center $(6, -8)$, Radius 10

Complete the square: $(x^2 - 6x + 9) + (y^2 + 8y + 16) = 9 + 16 \Rightarrow (x - 3)^2 + (y + 4)^2 = 25$. Center is $(3, -4)$ and radius is $\sqrt{25} = 5$.

45 [Challenging] • Target: Asymptotes and Branches of Hyperbolas • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Find the equations of the asymptotes for the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $y = \pm\frac{3}{4}x$ ✓

b. $y = \pm\frac{4}{3}x$

c. $y = \pm\frac{9}{16}x$

d. $y = \pm\frac{16}{9}x$

For a horizontal hyperbola centered at origin $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, asymptotes are $y = \pm\frac{b}{a}x$. Here $a = 4, b = 3$, giving $y = \pm\frac{3}{4}x$.

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